Revised Date	Reversion	Modified By	Note
2023/4/12	R0.1	Louis Chang	Initial release
2023/4/21	R1.0	Louis Chang	Initial release
2023/6/15	R2.0	Louis Chang	1. Depop R26, Q1, R27, R15, R6, R28, Q2, R30, R18, R24 2. Populate R12, R14, R13, R17 to 0ohm 3. Change C26, C27 to 68pF 4. Populate R25 to 17.4K ohm 5. Change R301 to 0 ohm 6. Change R2, R3 to 0 ohm 7. Remove jumper on LP_MODE pin on DVK 8. Change pin3 of SW2 to net BT_UART_CTS 9. Connect J3 pin2 to RP2040 GPIO21 (pin32) 10. add jumper on J3 pin9-pin10
2023/6/27	R3.0	Louis Chang	1. Swap net pin20 of J2 and pin4 of J3 2. Add one color LED oconnect to IF820 pin 33 (P27) 3. Change Y1 from Epson/FC-135 32.7680KA-AG5 to Siward/XTL721-S349-005
2023/7/6	R3.1	Louis Chang	1. Change all related net names from CY_ROLE to CP_ROLE
2023/7/27	R4.0	Louis Chang	1. Swap net of BT_UART_TXD_MikroE and BT_UART_RXD_MikroE 2. Add SB2(closed) between UART_1_RX_RP2040 and P_UART_TXD_20820 3. Add SB3(closed) between UART_1_TX_RP2040 and P_UART_RXD_20820
2024/6/26	R4.1	AL	Replace brand to Ezurio

LBL1
Ezurio, 160-00032

PCB1
Ezurio, 750-03405

Approvals	Date
Drawn: AL	4/12/2023
Checked: LC	4/21/2023
Approved: AC	4/21/2023

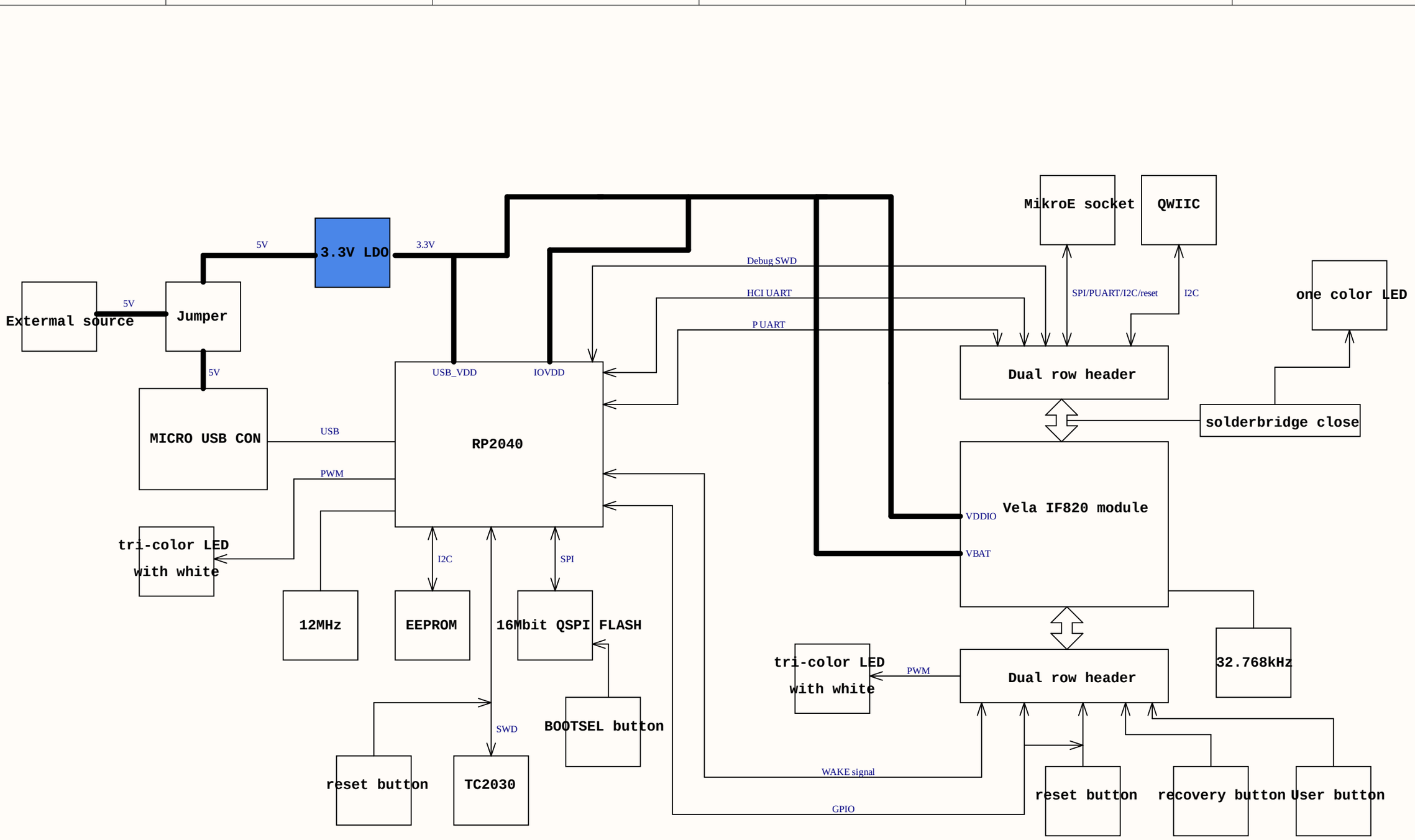
3F-1, No.145, Xianzheng 9th Rd., Zhubei City
Hsinchu County 30251, Taiwan (R.O.C.)
Phone: +886-3-5529695
www.ezurio.com

Title:
Schematic, Vela IF820 Module Dev Board, Integrated Antenna

Sheet Name:	Project Number:	Assembly Number:	Drawing Number:	Rev.
History	941-03405	940-00297	CS-DOC-000013039	**4.1**

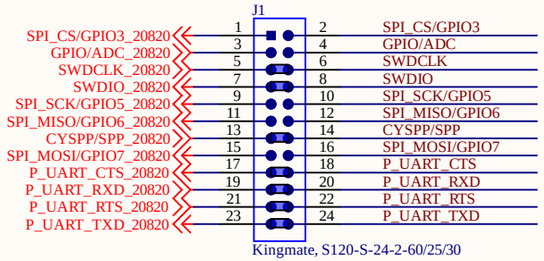
File:	Customer:	Last Modified:	Size:	Sheet:
CS-DOC-000013039[History].SchDoc	Internal	06/26/2024 6:26:00 PM	B	1 of 6

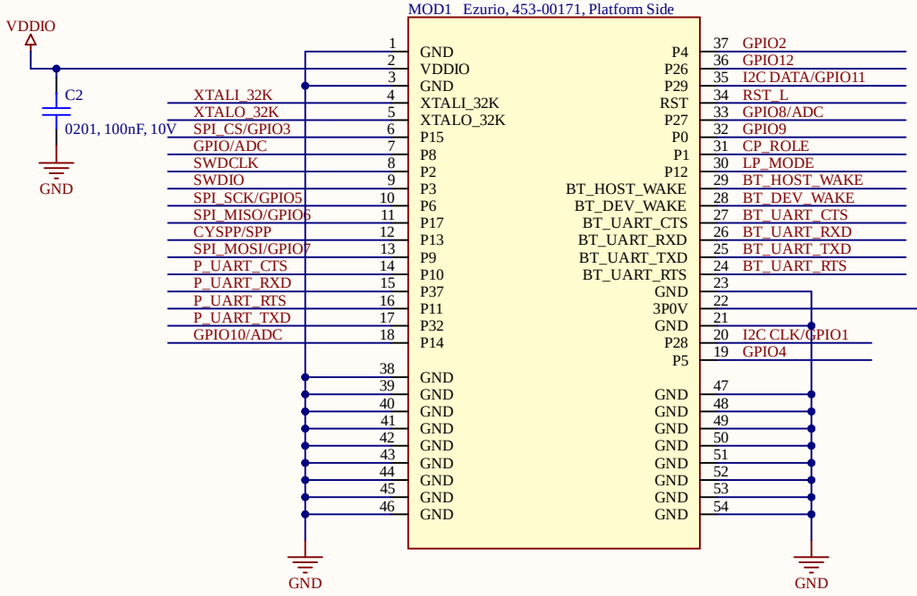
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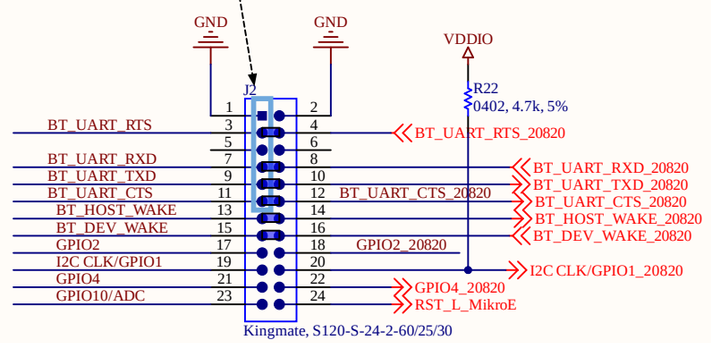
Title: Schematic, Vela IF820 Module Dev Board, Integrated Antenna					
Sheet Name: *Block Diagram*	Project Number: 941-03405	Assembly Number: 940-00297	Drawing Number: CS-DOC-000013039	Rev. **4.1**	
File: CS-DOC-000013039[Block Diagram].SchDoc	Customer: Internal	Last Modified: 06/26/2024 6:26:00 PM	Size: B	Sheet: 2 of 6	

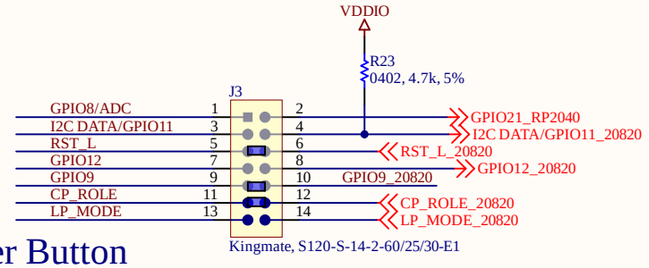


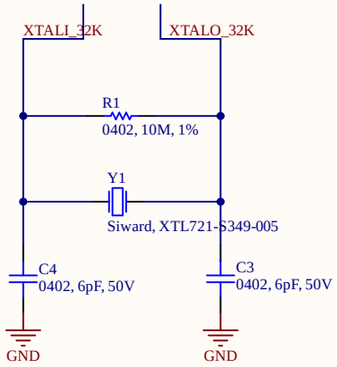


Pin sequence comply with FTDI cable (USB to TTL 232)

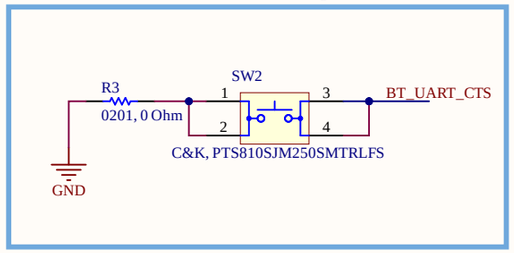
Please be sure to disconnect VCC of FTDI cable, and check FTDI voltage (3.3/1.8V) comply with module VDDIO



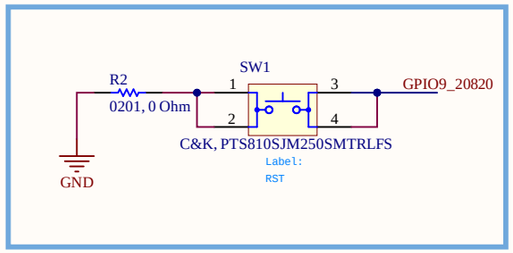


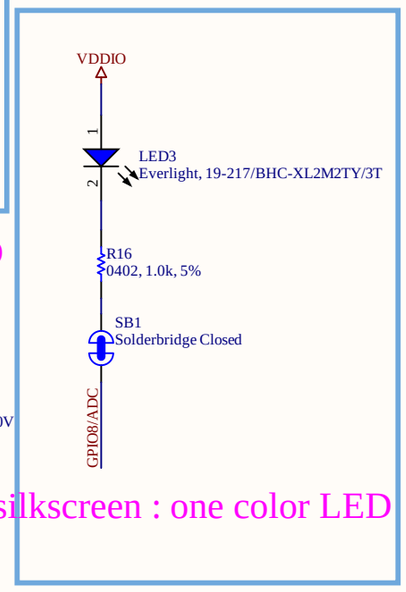


20820 Recovery button

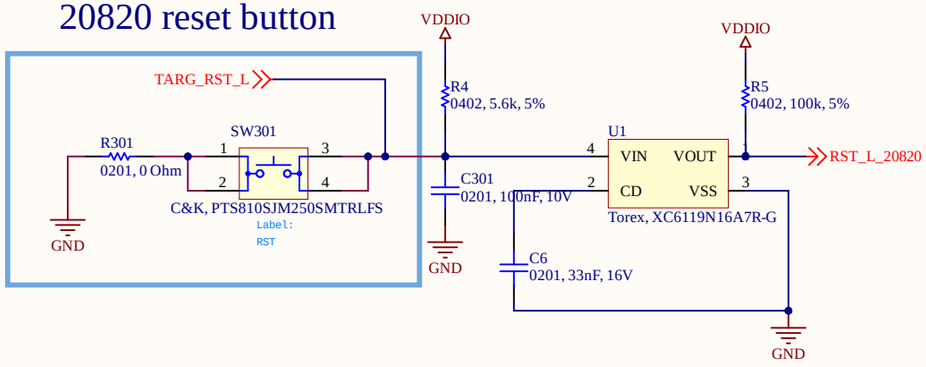


20820 User Button

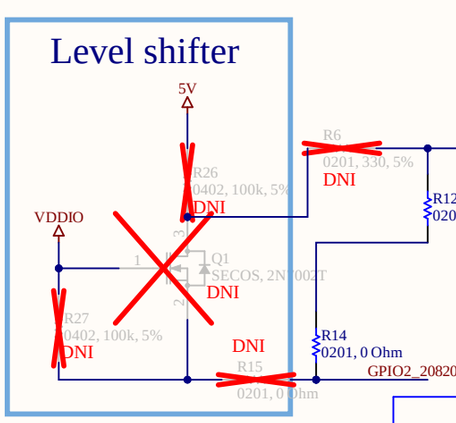


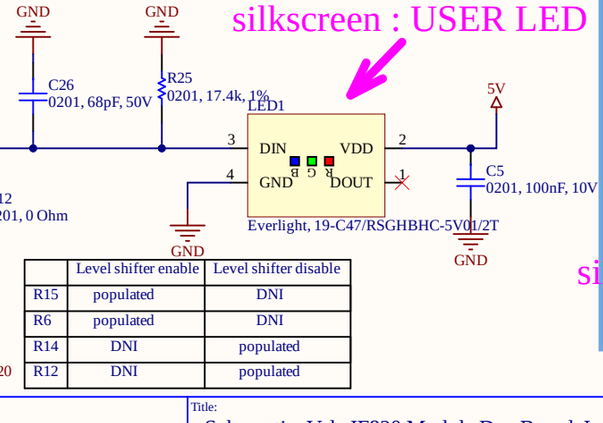


20820 reset button



Level shifter





silkscreen : USER LED

	Level shifter enable	Level shifter disable
R15	populated	DNI
R6	populated	DNI
R14	DNI	populated
R12	DNI	populated

silkscreen : one color LED

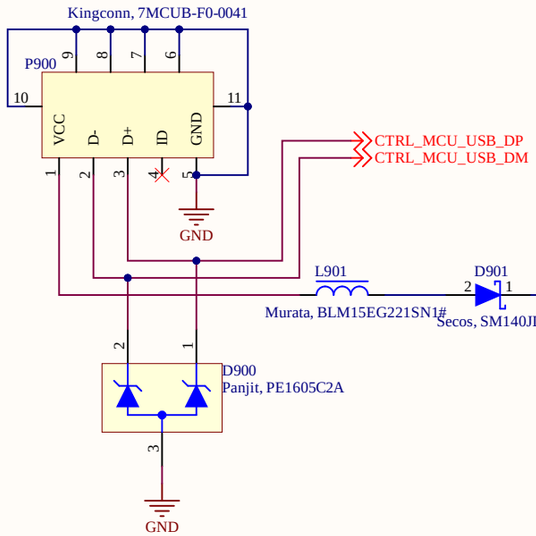
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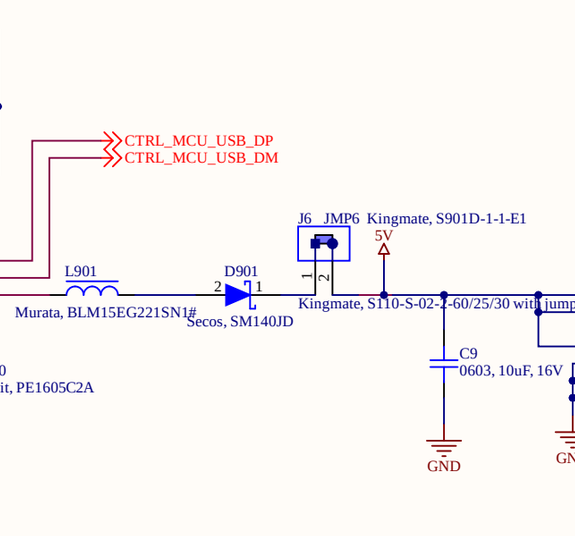
Title: Schematic, Vela IF820 Module Dev Board, Integrated Antenna

Sheet Name:	Project Number:	Assembly Number:	Drawing Number:	Rev.
Vela IF820	941-03405	940-00297	CS-DOC-000013039	4.1

File:	Customer:	Last Modified:	Size:	Sheet:
CS-DOC-000013039[Vela IF820-SA].SchDoc	Internal	06/26/2024 6:26:00 PM	B	3 of 6

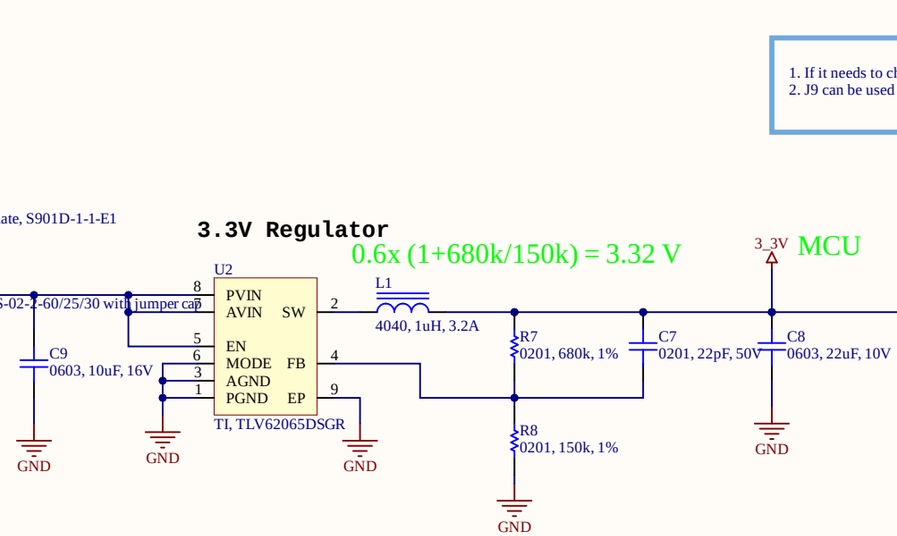
Debug USB Connection



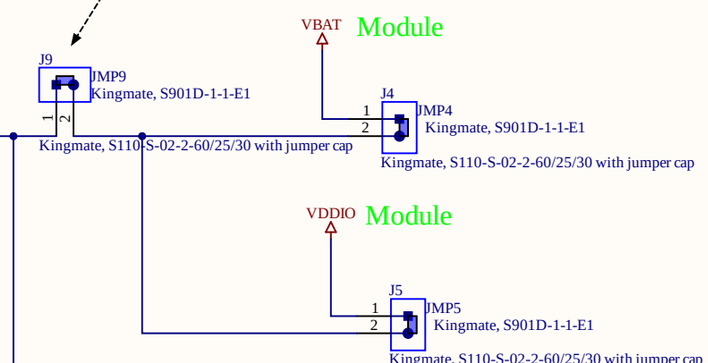


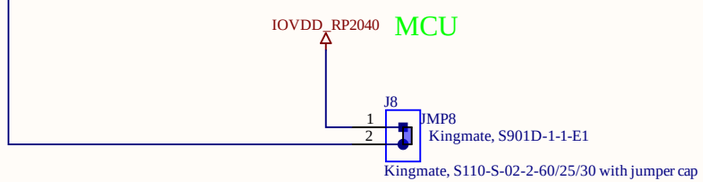
3.3V Regulator

0.6x (1+680k/150k) = 3.32 V



1. If it needs to change voltage of module (both VBAT and VDDIO), you can use J9
2. J9 can be used to current measured if VBAT and VDDIO are the same voltage

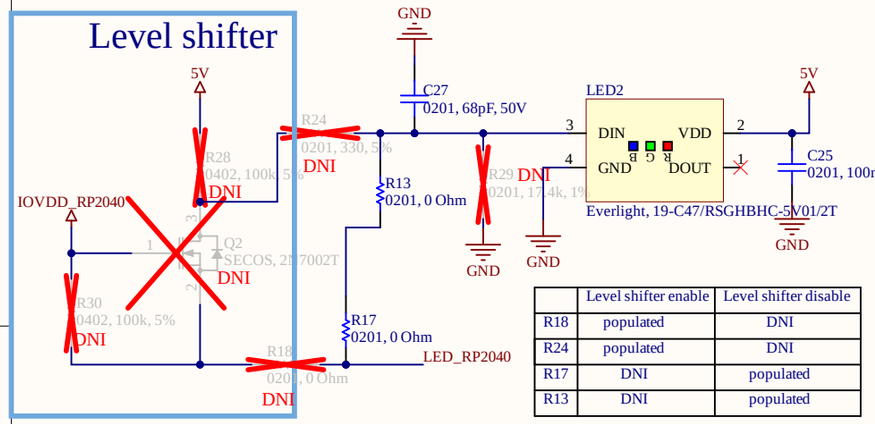




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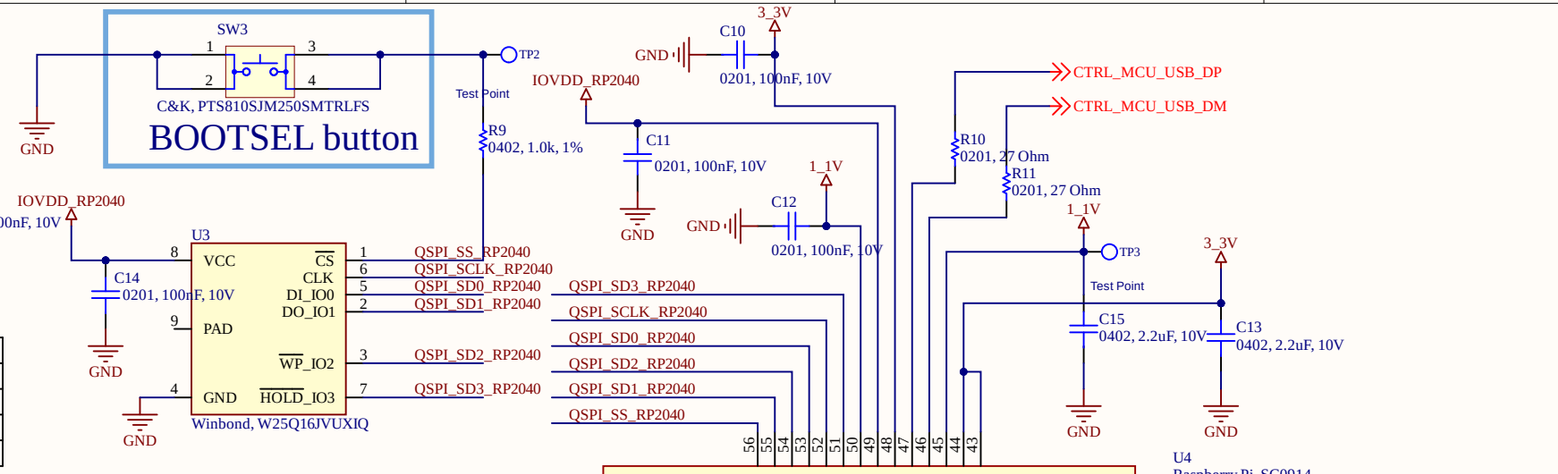
Title: Schematic, Vela IF820 Module Dev Board, Integrated Antenna					
Sheet Name: ***Board Power***	Project Number: 941-03405	Assembly Number: 940-00297	Drawing Number: CS-DOC-000013039	Rev. **4.1**	
File: CS-DOC-000013039[Board Power].SchDoc	Customer: Internal	Last Modified: 06/26/2024 6:26:00 PM	Size: B	Sheet: 4 of 6	

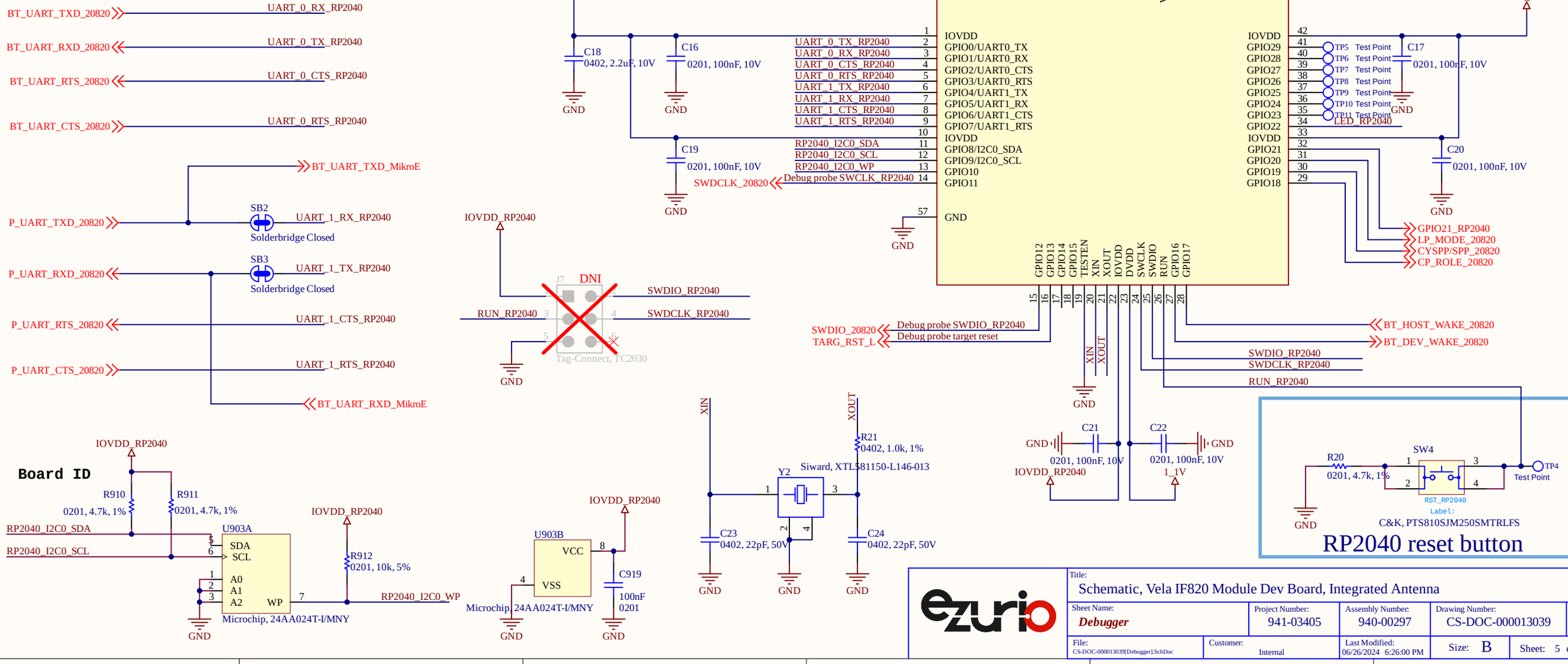
Level shifter



	Level shifter enable	Level shifter disable
R18	populated	DNI
R24	populated	DNI
R17	DNI	populated
R13	DNI	populated

BOOTSEL button

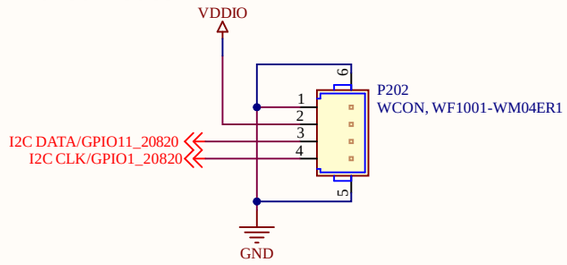




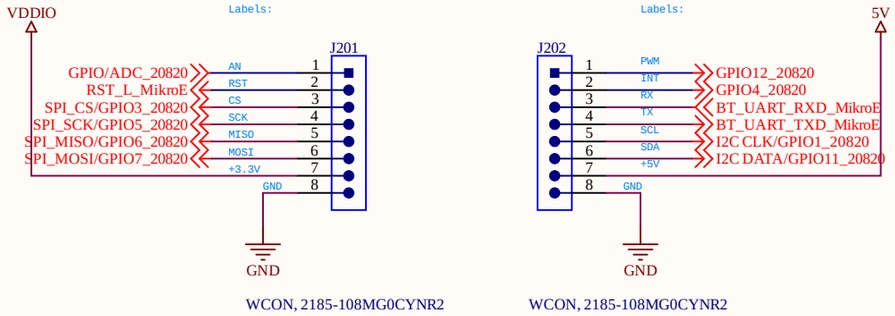
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Title: Schematic, Vela IF820 Module Dev Board, Integrated Antenna				
Sheet Name: ***Debugger***	Project Number: 941-03405	Assembly Number: 940-00297	Drawing Number: CS-DOC-000013039	Rev. **4.1**
File: CS-DOC-000013039[Debugger].SchDoc	Customer: Internal	Last Modified: 06/26/2024 6:26:00 PM	Size: B	Sheet: 5 of 6

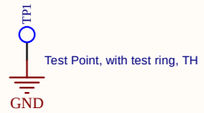
Qwiic Connector



MikroE Socket



Screw holes for production fixture



MH2
Mounting Hole, Hole 1.524mm, Dia 1.524mm

MH1
Mounting Hole, Hole 1.524mm, Dia 1.524mm

MH4
Mounting Hole, Hole 1.524mm, Dia 1.524mm

MH3
Mounting Hole, Hole 1.524mm, Dia 1.524mm

MH6
Mounting Hole, Hole 3.2mm, Dia 6mm

MH5
Mounting Hole, Hole 3.2mm, Dia 6mm

MH8
Mounting Hole, Hole 3.2mm, Dia 6mm

MH7
Mounting Hole, Hole 3.2mm, Dia 6mm

MH10
Mounting Hole, Hole 3.2mm, Dia 6mm

MH9
Mounting Hole, Hole 3.2mm, Dia 6mm

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Title: Schematic, Vela IF820 Module Dev Board, Integrated Antenna

Sheet Name:	Project Number:	Assembly Number:	Drawing Number:	Rev.
User Interface	941-03405	940-00297	CS-DOC-000013039	**4.1**

File:	Customer:	Last Modified:	Size:	Sheet:
CS-DOC-000013039[User Interface].SchDoc	Internal	06/26/2024 6:26:01 PM	B	6 of 6